

Hasan Abbani

Data Analytics | Artificial Intelligence

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Profile

MSc student in Computational Science with experience in data analytics, machine learning, and technical research. Skilled in Python, SQL, Power BI, and statistical modeling, with an interest in AI and digital transformation.

Technical Skills

Programming: Python, R, C++, Java (OOP)

Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn; supervised and unsupervised learning (regression, classification, clustering, tree-based models, neural networks)

Data Analysis & Visualization: Excel for data analysis and reporting, Power BI (basic dashboarding and reporting), exploratory data analysis, feature engineering, KPI reporting, data visualization using Matplotlib, Microsoft PowerPoint (research presentations and reporting)

Statistical Learning & Modeling: Regularization methods (Ridge, Lasso), cross-validation and model selection, bias–variance analysis, dimensionality reduction (PCA, PCR)

Deep Learning & AI: PyTorch, transformer architectures, attention mechanisms, generative AI concepts

APIs & Software Tools: REST APIs, Git/GitHub, external data integration

Reinforcement Learning: Q-learning, Deep Q-Networks (DQN)

Databases & Data Management: SQL, MySQL, database querying, joins, aggregation, filtering, ETL pipelines

Algorithms & Optimization: Graph algorithms; gradient-based optimization

Languages: English (Fluent), Arabic (Native), and German (A1)

Education and Awards

MSc Computational Science – Data Analytics Track

American University of Beirut (AUB)

Aug 2025 – June 2027 (Expected)

Awarded the Graduate Fellowship and Assistantship, a full (100%) merit-based scholarship

Beirut Arab University (BAU) — *Beirut, Lebanon*

Bachelor's Degree in Physics, Graduated with Honors

Sep 2022 – May 2025

Minor in Computer Science

CGPA: **3.67 / 4.0**

Received 80% Merit-Based Scholarship

Professional Experience

Teaching Assistant at AUB– Data Science Lab

Sep 2025 – Dec 2025

- Assisted in undergraduate data science labs, supporting Python/R-based data analysis.
- Guided students in data visualization, exploratory analysis, and model interpretation.
- Evaluated assignments and provided targeted feedback on computational workflows.

Internship, Materials Science Research Laboratory at BAU

Sep 2023 – Jun 2024

- Gained familiarity with academic writing standards and the research process.

- Obtained hands-on experience using materials science laboratory equipment.
- Contributed to a study on the dielectric properties of a ferrite nanocomposite.

Tutor, Private Tutoring

Jan 2023 – Jun 2024

- Tutored mathematics, physics, and biophysics students at secondary and undergraduate levels.

Team Leader, Bariq

Nov 2021 – Dec 2022

- Led and coordinated two photography teams (around 40 members), delivered training sessions, and organized participation in community events.

Achievements

1st Place, Lebanon's First Quantum Computing Hackathon – AUB

American University of Beirut | April 2025

- Organized by **CAMS, AUB Quantum Club**, and **CERN's Open Quantum Institute** in partnership with **NYU Abu Dhabi**.
- Worked in a team to design **Q-TES**, a Quantum Traffic Enhancer System using **QAOA** to optimize urban traffic light scheduling. [\[GitHub\]](#)
- The project aimed to reduce **CO₂ emissions**, aligning with the **UN Sustainable Development Goals**.

Projects

GitHub

- **ETL & Machine Learning Pipeline for Energy Data Analytics:** Built an end-to-end Python pipeline to automate data extraction, cleaning, transformation, and integration from external APIs. Generated weekly KPIs, visualizations, and machine learning forecasts to identify trends and anomalies in energy consumption data. [Visit the Project](#)

Senior Project

- **Neural Networks for Predicting the Bulk Modulus of Materials:** Built a neural network in PyTorch to predict the bulk modulus of materials. Used about 11000 materials with both their experimental and theoretical features, in addition to engineered features. Achieved an R^2 score of 0.91. The project received an **A+** and high praise from professors.

Google Colab

- **Candlestick-Based Time-Series Analysis:** Engineered geometric candlestick features and evaluated ML models to assess predictive signal in short-term financial time series. [Visit the Project](#)
- **Pulsar Classification Project:** Applied advanced ML techniques, visualization, and data augmentation to classify the light of pulsars. [Visit the Project](#)

Kaggle

- **A Statistical Approach to the Collatz Conjecture:** Used statistical analysis and visualization to make predictions about the behavior of numbers in the conjecture. [Visit the Project](#)

Volunteer Work

Member at Large, Quantum Club – AUB

Sep 2025 – Present

Executive Vice President, Astronomy Club – BAU

Sep 2024 – Aug 2025

Organizer, Science Fair at Beirut Arab University

March 2023